



1. Hotelling's T^2 Test for a Mean Vector

```
In [97...] X <- matrix(  
  c(2, 12,  
    8, 9,  
    6, 9,  
    8, 10),  
  nrow = 4,  
  byrow = TRUE  
)  
  
print(X)
```

```
      [,1] [,2]  
[1,]     2    12  
[2,]     8     9  
[3,]     6     9  
[4,]     8    10
```

1.a

```
In [98...] n <- nrow(X)  
p <- ncol(X)  
  
xbar <- colMeans(X)  
cat("The sample mean vector is:")  
xbar  
  
S <- cov(X)  
  
mu0 <- c(7, 11)  
  
d <- xbar - mu0  
  
T2 <- n * t(d) %*% solve(S) %*% d  
  
cat("Hotelling's T^2 is:")  
T2
```

The sample mean vector is:
6 10

Hotelling's T^2 is:
A matrix: 1
× 1 of type
dbl
13.63636

1.b

Under the null hypothesis, T^2 transforms into an F -distribution where:

$$T^2 \sim \frac{p(n-1)}{n-p} F_{\{p, n-p\}}$$

Substituting $n=4$ and $p=2$, we get

$$T^2 \sim \frac{2(3)}{2} F_{\{2, 2\}} = 3 F_{\{2, 2\}}$$

1.c

```
In [99... crit_F <- qf(0.95, p, n - p)
crit_T2 <- (p * (n - 1) / (n - p)) * crit_F

cat("Critical T^2 value is:")
crit_T2
```

Critical T^2 value is:

56.99999999999999

Since Critical T^2 value is greater than T^2 value, this means we fail to reject the null hypothesis. This means that there is not enough evidence to conclude that the population mean vector differs from $\mu^{\prime} = [7, 11]$ at the 5% significance level.

This means that the sample aligns with the theorized mean vector

$\mu^{\prime} = [7, 11]$.

2. Simultaneous Confidence Intervals and Regions for Grizzly Bear Body Measurements

```
In [10... n <- 61
p <- 6

xbar <- c(
  Weight = 95.52,
  Body_Length = 164.38,
  Neck = 55.69,
  Girth = 93.39,
  Head_Length = 17.98,
  Head_Width = 31.13
)
```

```
xbar

S <- matrix(
  c(
    3266.46, 1343.97, 731.54, 1175.50, 162.68, 238.37,
    1343.97, 721.91, 324.25, 537.35, 80.17, 117.73,
    731.54, 324.25, 179.28, 281.17, 39.15, 56.80,
    1175.50, 537.35, 281.17, 474.98, 63.73, 94.85,
    162.68, 80.17, 39.15, 63.73, 9.95, 13.88,
    238.37, 117.73, 56.80, 94.85, 13.88, 21.26
  ),
  nrow = 6,
  byrow = TRUE
)

S
```

Weight: 95.52 Body_Length: 164.38 Neck: 55.69 Girth: 93.39
Head_Length: 17.98 Head_Width: 31.13

A matrix: 6 × 6 of type dbl

3266.46	1343.97	731.54	1175.50	162.68	238.37
1343.97	721.91	324.25	537.35	80.17	117.73
731.54	324.25	179.28	281.17	39.15	56.80
1175.50	537.35	281.17	474.98	63.73	94.85
162.68	80.17	39.15	63.73	9.95	13.88
238.37	117.73	56.80	94.85	13.88	21.26

2.a

```
In [10... chi_val <- qchisq(0.95, df = p)
margin <- sqrt(
  chi_val * diag(S) / n
)

CI <- cbind(
  Lower = xbar - margin,
  Upper = xbar + margin
)

cat("95% Simultaneous Confidence Intervals for the six population mean
body measurements:\n")
CI
```

95% Simultaneous Confidence Intervals for the six population mean
body measurements:

A matrix: 6 × 2 of type dbl

	Lower	Upper
Weight	69.55347	121.48653
Body_Length	152.17278	176.58722
Neck	49.60667	61.77333
Girth	83.48823	103.29177
Head_Length	16.54687	19.41313
Head_Width	29.03513	33.22487

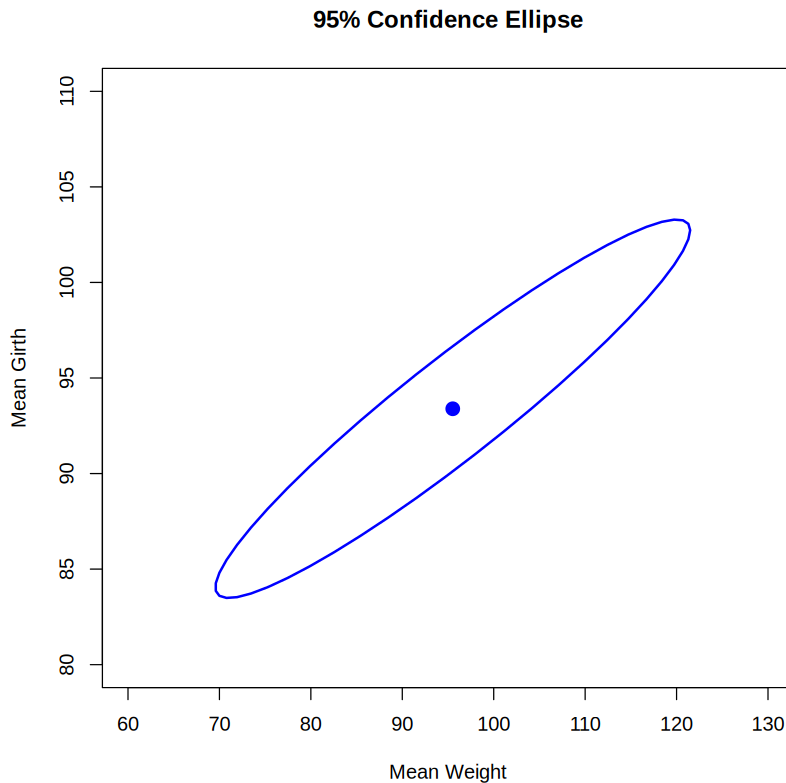
2.b

```
In [10... install.packages("car")
library(car)
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

```
In [10... sub_S <- S[c(1,4), c(1,4)]
sub_xbar <- xbar[c(1,4)]

plot(0, 0, type="n", xlim=c(60, 130), ylim=c(80, 110),
     xlab="Mean Weight", ylab="Mean Girth", main="95% Confidence Ellipse"
     ellipse(sub_xbar, sub_S/n, radius=sqrt(chi_val), col="blue")
```



The ellipse is centered at around (95.52,93.38), which is inline with the calculated mean for weight and girth respectively. The ellipse represents the probable combinations of the population mean vector for weight and girth simultaneously at the 95% confidence level.

2.c

```
In [10... m <- 6

t_bonferroni <- qt(1 - 0.05/(2*m), df = n-1)
bonferroni_margin <- t_bonferroni * sqrt(diag(S)/n)
bonferroni_CI <- cbind(Lower = xbar - bonferroni_margin, Upper = xbar + b

cat("Bonferroni CIs for the six means in part (a):")
bonferroni_CI
```

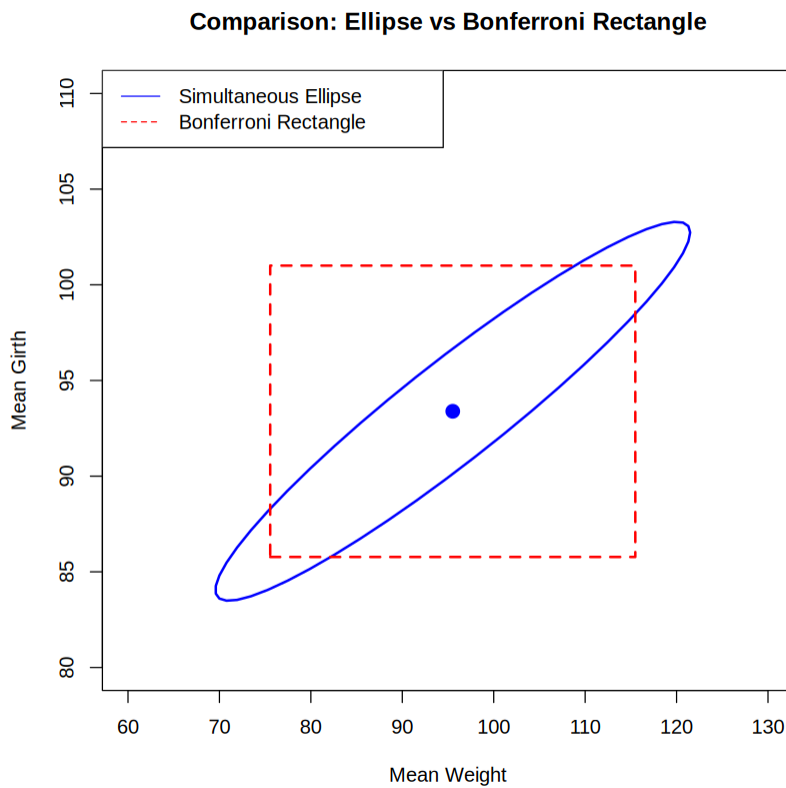
Bonferroni CIs for the six means in part (a):

A matrix: 6 × 2 of type dbl

	Lower	Upper
Weight	75.55331	115.48669
Body_Length	154.99339	173.76661
Neck	51.01229	60.36771
Girth	85.77614	101.00386
Head_Length	16.87801	19.08199
Head_Width	29.51917	32.74083

2.d

```
In [10... plot(0, 0, type="n", xlim=c(60, 130), ylim=c(80, 110),
      xlab="Mean Weight", ylab="Mean Girth",
      main="Comparison: Ellipse vs Bonferroni Rectangle")
ellipse(sub_xbar, sub_S/n, radius=sqrt(chi_val), col="blue")
rect(bonferroni_CI[1,1], bonferroni_CI[4,1], bonferroni_CI[1,2], bonferro
legend("topleft", legend=c("Simultaneous Ellipse", "Bonferroni Rectangle"
      col=c("blue", "red"), lty=c(1, 2))
```



The Bonferroni confidence rectangle was constructed using the separate

Bonferroni confidence intervals for mean weight and mean girth. Meanwhile the ellipse uses the calculated covariance, which considers the weight and girth together. This means that the Bonferroni confidence rectangle and simultaneous confidence ellipse have different ways of calculating the joint uncertainty in the mean weight and mean girth.

For the graph while there is some overlap, there are noticeable differences. To start, the ellipse is longer compared to the rectangle, both of its x-axis and y-axis extend past the rectangle. Meanwhile, the upper-left and bottom-right corners of the rectangle has values which the ellipse doesn't. Finally, the ellipse is tilted upward, which shows the positive correlation between the mean weight and mean girth. This relationship wasn't captured by the Bonferroni confidence rectangle since the rectangle calculated the intervals separately compared to the ellipse which included the covariance.

2.e

```
In [10... m_new <- 7
t_bonferroni_new <- qt(1 - 0.05/(2*m_new), df = n-1)

diff_mean <- xbar[6] - xbar[5]
diff_var <- S[6,6] + S[5,5] - 2*S[5,6]
diff_se <- sqrt(diff_var / n)
diff_CI <- c(diff_mean - t_bonferroni_new * diff_se, diff_mean + t_bonfer

cat("Bonferroni CI for Difference (Head Width - Length):")
diff_CI
```

Bonferroni CI for Difference (Head Width - Length):

Head_Width: 12.4875653260697 **Head_Width:** 13.8124346739303

Population mean Head_Length is 17.98 and population mean Head_Width is 31.13. The estimated difference between them is $31.13 - 17.98 = 13.15$. Using $m = 7$, the 95% Bonferroni CI is (12.4875653260697, 13.8124346739303). This means that we are 95% confident that the difference between the population mean head width and population mean head length is between approximately 12.49 cm and 13.81 cm. Since this interval is positive, the population population mean head width is estimated to be greater than the population mean head length.

3

```
In [10... x1 <- c(0.48, 40.53, 2.19, 0.55, 0.74, 0.66, 93.00, 0.37, 0.22)
x2 <- c(12.57, 73.68, 11.13, 20.03, 20.29, 0.78, 4.64, 0.43, 1.08)

peru_data <- data.frame(
  cr = x1,
  st = x2
)

n <- nrow(peru_data)
p <- ncol(peru_data)
xbar <- colMeans(peru_data)
S <- cov(peru_data)

peru_data
cat("Column Means (X-bar):\n")
xbar

cat("Covariance (S):\n")
S
```

A data.frame: 9
× 2

	cr	st
	<dbl>	<dbl>
	0.48	12.57
	40.53	73.68
	2.19	11.13
	0.55	20.03
	0.74	20.29
	0.66	0.78
	93.00	4.64
	0.37	0.43
	0.22	1.08

Column Means (X-bar):
cr: 15.415555555555556 **st:** 16.07

Covariance (S):
A matrix: 2 × 2 of type dbl

	cr	st
cr	1019.9280	155.6962
st	155.6962	527.8493

3.a

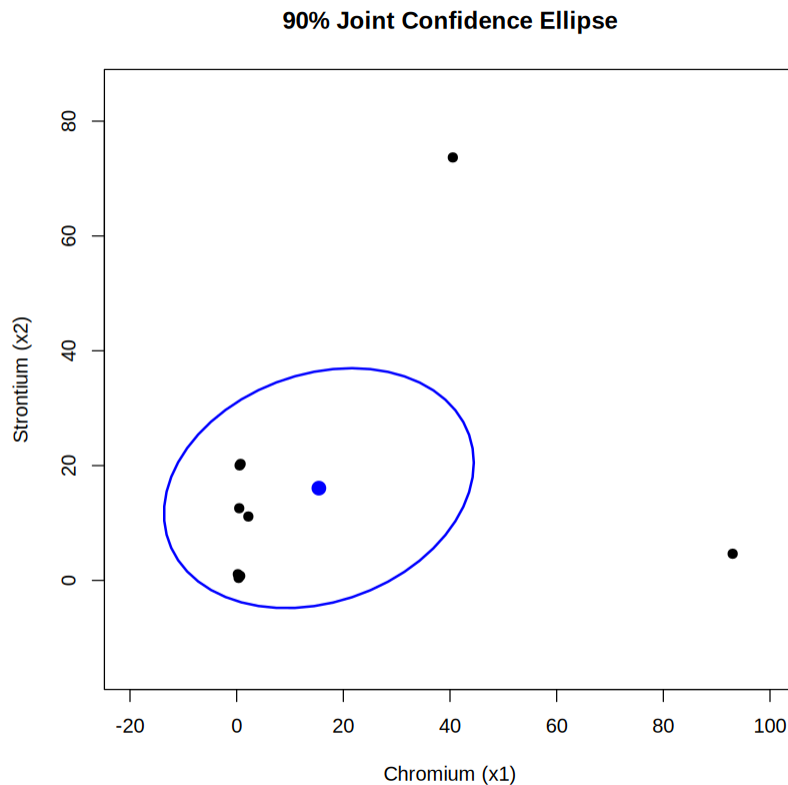
```

In [10... F_crit <- qf(0.90, p, n - p)
crit_val<- ((n - 1) * p / (n - p)) * F_crit

plot(peru_data$cr, peru_data$st,
     xlim=c(-20, 100),ylim=c(-15, 85),
     xlab="Chromium (x1)", ylab="Strontium (x2)",
     pch=19, main="90% Joint Confidence Ellipse")

ellipse(xbar, S/n, radius=sqrt(crit_val), col="blue")

```



3.b

```

In [10... margin <- sqrt(crit_val) * sqrt(diag(S)/n)
CI <- cbind(Lower = xbar - margin, Upper = xbar + margin)
cat("Simultaneous 90% CIs:")
CI

test_mu <- c(0.30, 10)
T2_test <- n * t(xbar - test_mu) %*% solve(S) %*% (xbar - test_mu)
cat("T2 for [.30, 10]:", T2_test, "\n")
cat("Critical Value:", crit_val, "\n")

```

Simultaneous 90% CIs:

A matrix: 2 × 2 of type dbl

	Lower	Upper
cr	-13.632217	44.46333
st	-4.826957	36.96696

T2 for [.30, 10]: 2.268901

Critical Value: 7.445582

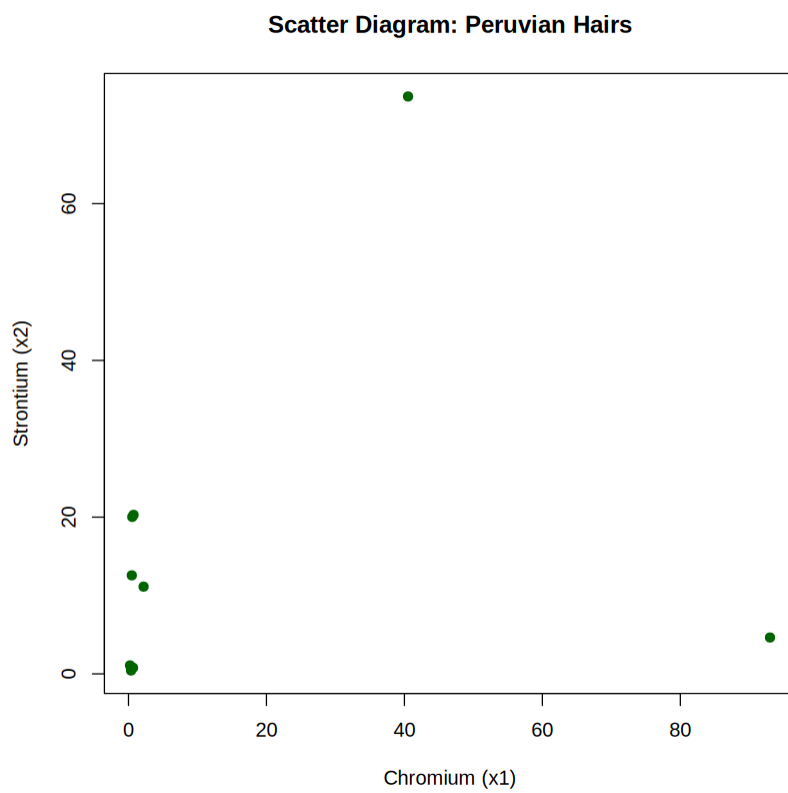
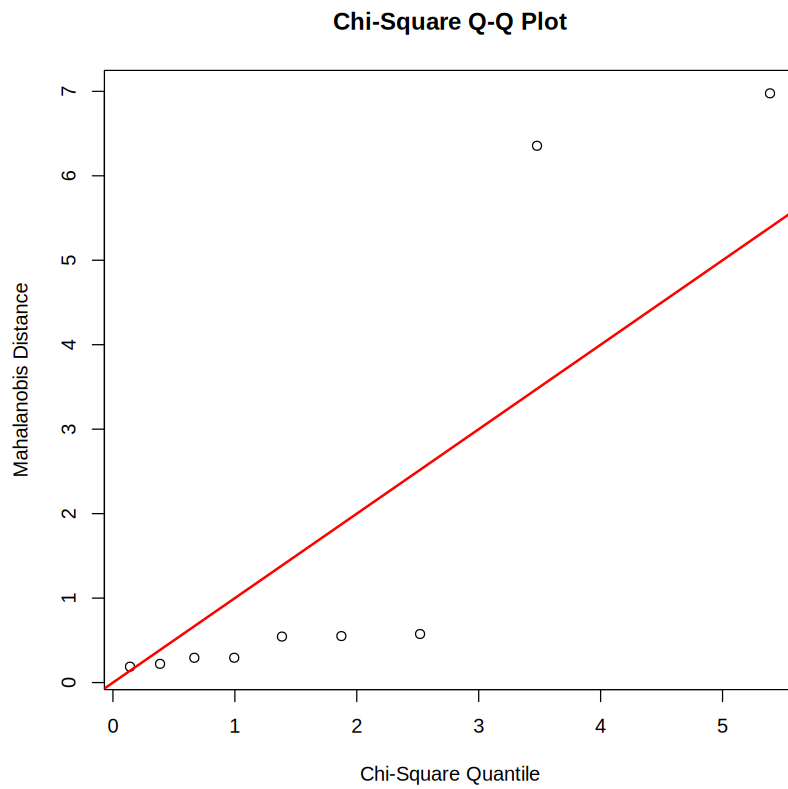
The simultaneous 90% confidence interval for strontium is (-4.826957,36.96696). Since 10 is inside that interval, this means that the data does not provide evidence against a population mean strontium level of 10 ppm.

For the proposed population mean vector [0.30,10]', T^2 is calculated to be 2.268901. Since the critical value is 7.445582, this means T^2 is well below the critical value. Thus [0.30,10]' is considered to be a plausible value for the population mean vector based on the 90% confidence region.

3.c

```
In [11... d2 <- mahalanobis(peru_data, xbar, S)
qqplot(qchisq(ppoints(n), df = p), d2,
       main = "Chi-Square Q-Q Plot", xlab="Chi-Square Quantile", ylab="Ma
abline(a = 0, b = 1, col="red", lwd=2)

plot(peru_data$cr, peru_data$st,
     main="Scatter Diagram: Peruvian Hairs",
     xlab="Chromium (x1)", ylab="Strontium (x2)", pch=19, col="darkgreen")
```



Chi-Square Q-Q Plot: For a bivariate normal distribution, points should closely

follow the red reference line. However, in the graph produced, the points are mostly below the reference line while the rightmost points have a large jump in value, meaning this is a heavy-tailed, non-normal distribution.

Scatter Diagram: For a bivariate normal distribution, points should produce a roughly cohesive, elliptical "cloud." However, in the graph produced, the data does not follow this shape. Majority of the points are tightly clustered near the origin (very low Chromium and Strontium), while distinct, extreme outliers exist far from the cluster in the origin. This confirms the result from the Q-Q Plot.

Since the data are shown to NOT be bivariate normally distributed, this assumption was violated. This means that the theoretical Hotelling's (T^2) distribution, which was used to create the confidence ellipse and simultaneous confidence intervals, may not be accurate or reliable. Thus the results from part (a) and (b) may be inaccurate and lead to unreliable inferences and analysis.

3.d

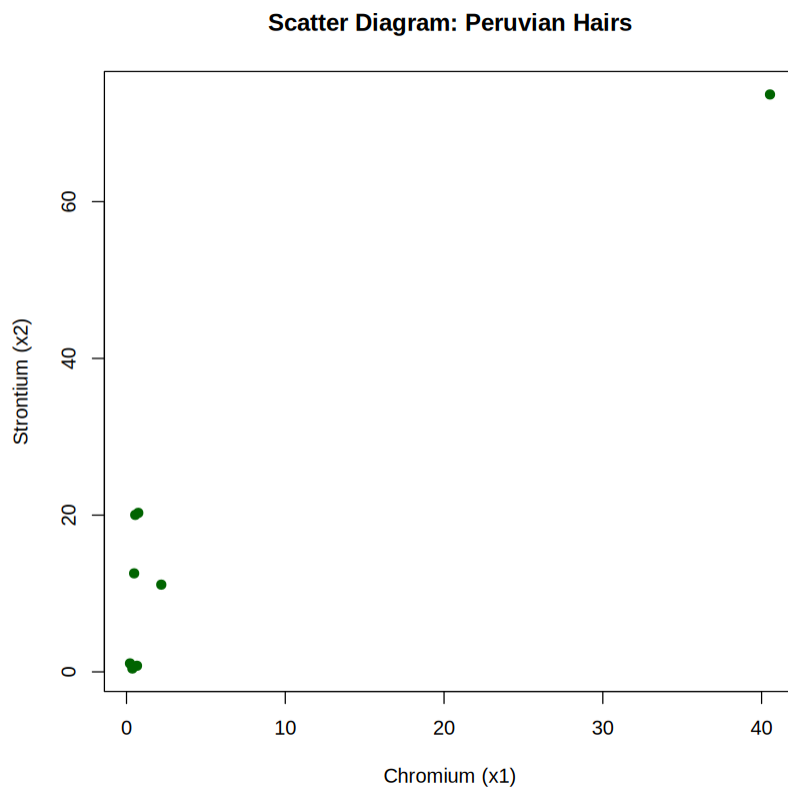
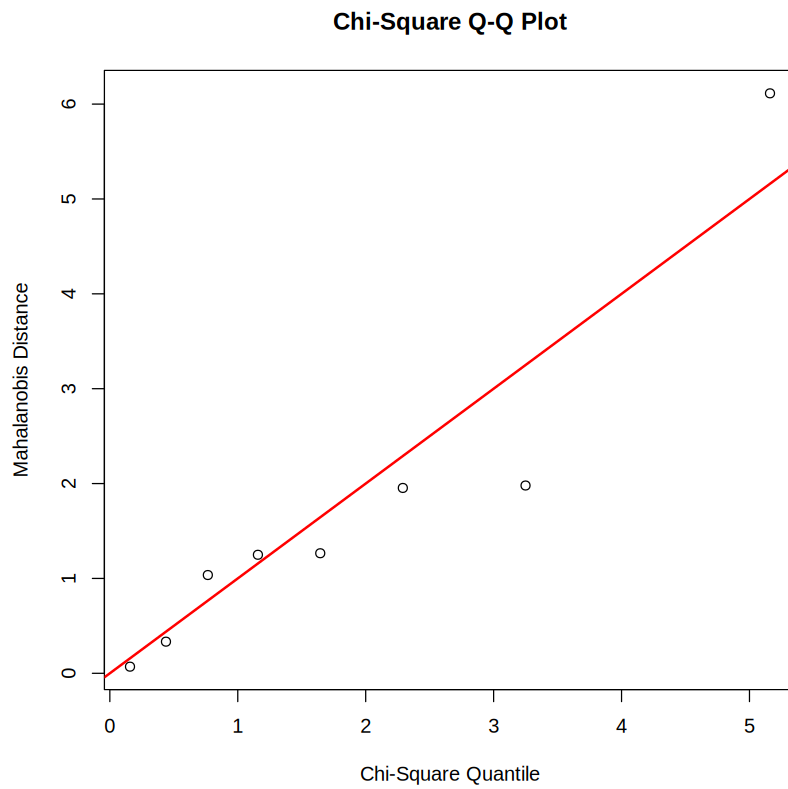
```
In [11... outlier <- which.max(d2)
clean_data <- peru_data[-outlier, ]
n_new <- nrow(clean_data)
xbar_new <- colMeans(clean_data)
S_new <- cov(clean_data)

crit_val_new <- ((n_new - 1) * p / (n_new - p)) * qf(0.90, p, n_new - p)
test_mu_new <- c(0.30, 10)
T2_test_new <- n_new * t(xbar_new - test_mu_c) %*% solve(S_new) %*% (xbar
cat("After outlier removal, New T2 for [.30, 10]:", T2_test_new, "\n")
cat("New Critical Value:", crit_val_new, "\n")
```

After outlier removal, New T2 for [.30, 10]: 1.381806
New Critical Value: 8.081043

```
In [11... d2 <- mahalanobis(clean_data, xbar_new, S_new)
qqplot(qchisq(ppoints(n_new), df = p), d2,
      main = "Chi-Square Q-Q Plot", xlab="Chi-Square Quantile", ylab="Ma
abline(a = 0, b = 1, col="red", lwd=2)

plot(clean_data$cr, clean_data$st,
      main="Scatter Diagram: Peruvian Hairs",
      xlab="Chromium (x1)", ylab="Strontium (x2)", pch=19, col="darkgreen"
```



After outlier removal, the new T^2 for $[.30, 10]$ is 1.381806 and the new

critical value is 8.081043. Since T^2 is still below critical value, so $[0.30, 10]'$ remains to be a plausible value for the population mean vector.

Q-Q Plot Assessment: The new plot shows that the points align slightly better with the red reference line than in the previous graph. However, it can still be seen that the point in the extreme upper tail deviates upward, which indicates that the data is still heavy-tailed and the assumption of bivariate normality is still questionable.

Scatter Diagram: The new scatter plot shows that while one extreme outlier was removed, the data still does not form the expected elliptical cloud of a bivariate normal distribution. The remaining points are still with one extreme point still visible far away from the cluster.

This means that bivariate normal distribution is still not satisfied, and the previous conclusion of $[0.30, 10]'$ being a plausible value should not be treated as completely accurate or reliable.

4

```
In [11... x1 = c(1232, 1115, 2205, 1897, 1932, 1612, 1598, 1804, 1752, 2067,
        2365, 1646, 1579, 1880, 1773, 1712, 1932, 1820, 1900, 2426,
        1558, 1470, 1858, 1587, 2208, 1487, 2206, 2332, 2540, 2322)

x2 = c(4175, 6652, 7612, 10914, 10850, 7627, 6954, 8365, 9469, 6410,
        10327, 7320, 8196, 9709, 10370, 7749, 6818, 9307, 6457, 10102,
        7414, 7556, 7833, 8309, 9559, 6255, 10723, 5430, 12090, 10072)

lumber_data <- data.frame(
  stiffness = x1,
  bending_strength = x2
)

n <- nrow(lumber_data)
p <- ncol(lumber_data)
xbar <- colMeans(lumber_data)
S <- cov(lumber_data)

lumber_data

cat("Column Means (X-bar):\n")
xbar

cat("Covariance (S):\n")
S
```

A data.frame: 30 × 2

stiffness **bending_strength**

<dbl>	<dbl>
1232	4175
1115	6652
2205	7612
1897	10914
1932	10850
1612	7627
1598	6954
1804	8365
1752	9469
2067	6410
2365	10327
1646	7320
1579	8196
1880	9709
1773	10370
1712	7749
1932	6818
1820	9307
1900	6457
2426	10102
1558	7414
1470	7556
1858	7833
1587	8309
2208	9559
1487	6255
2206	10723
2332	5430
2540	12090
2322	10072

Column Means (X-bar):

stiffness: 1860.5 **bending_strength:** 8354.133333333333

Covariance (S):

A matrix: 2 × 2 of type dbl

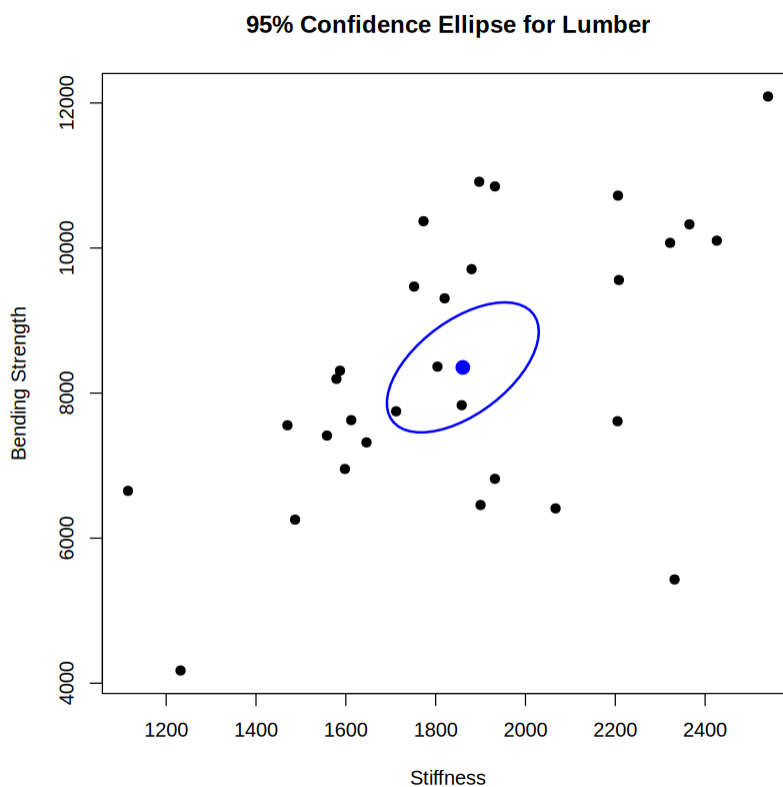
	stiffness	bending_strength
stiffness	124054.7	361620.4
bending_strength	361620.4	3486333.2

4.a

```
In [11... F_crit <- qf(0.95, p, n - p)
crit_val <- ((n - 1) * p / (n - p)) * F_crit

plot(lumber_data$stiffness, lumber_data$bending_strength, xlab="Stiffness",
     pch=19, main="95% Confidence Ellipse for Lumber")

ellipse(xbar, S/n, radius=sqrt(crit_val), col="blue")
```



4.b

```
In [11... mu <- c(2000, 10000)
T2 <- n * t(xbar - mu) %*% solve(S) %*% (xbar - mu)
```

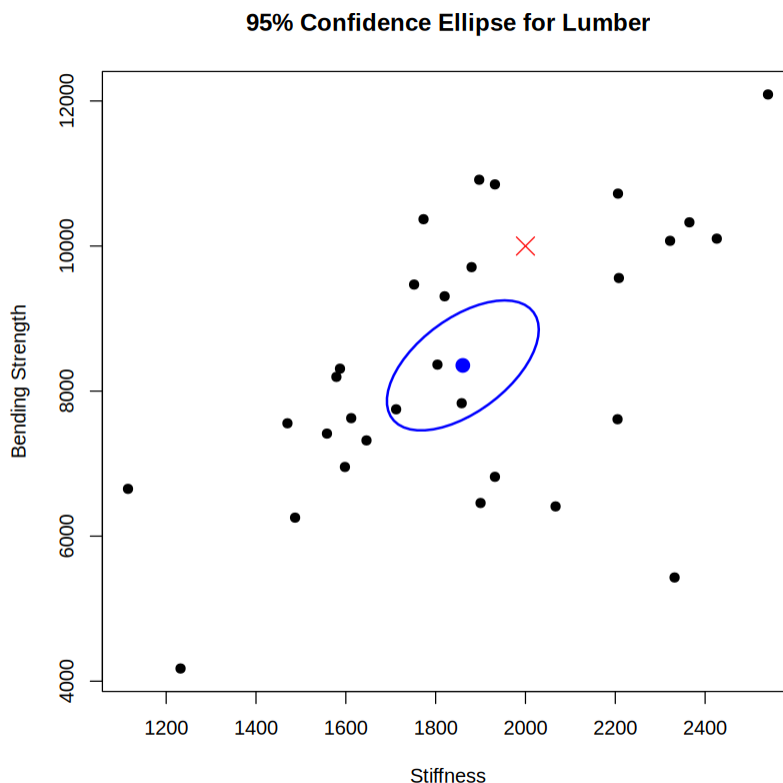
```

plot(lumber_data$stiffness, lumber_data$bending_strength, xlab="Stiffness",
     pch=19, main="95% Confidence Ellipse for Lumber")
ellipse(xbar, S/n, radius=sqrt(crit_val), col="blue")
points(2000, 10000, col="red", pch=4, cex=2)

cat("T2 for [2000, 10000]:", T2, "\n")
cat("Critical value:", crit_val, "\n")

```

T2 for [2000, 10000]: 23.64778
Critical value: 6.91937



The proposed typical population mean vector is [2000,10000]' which has $T^2 = 23.64778$. Meanwhile the critical value for the 95% confidence region is 6.91937. Since $23.64778 > 6.91937$, this means that [2000,10000]' lies outside the 95% confidence region. Therefore [2000,10000]' is not consistent with the observed sample at the 95% confidence level.

The graph confirms this as the [2000,10000]' point (indicated by the red x), lies outside the confidence ellipse.

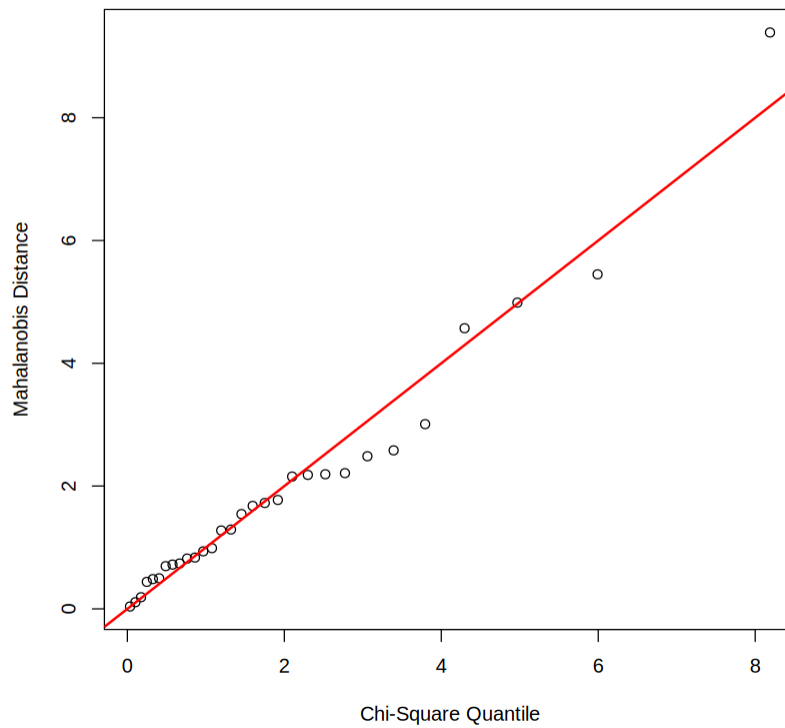
4.c

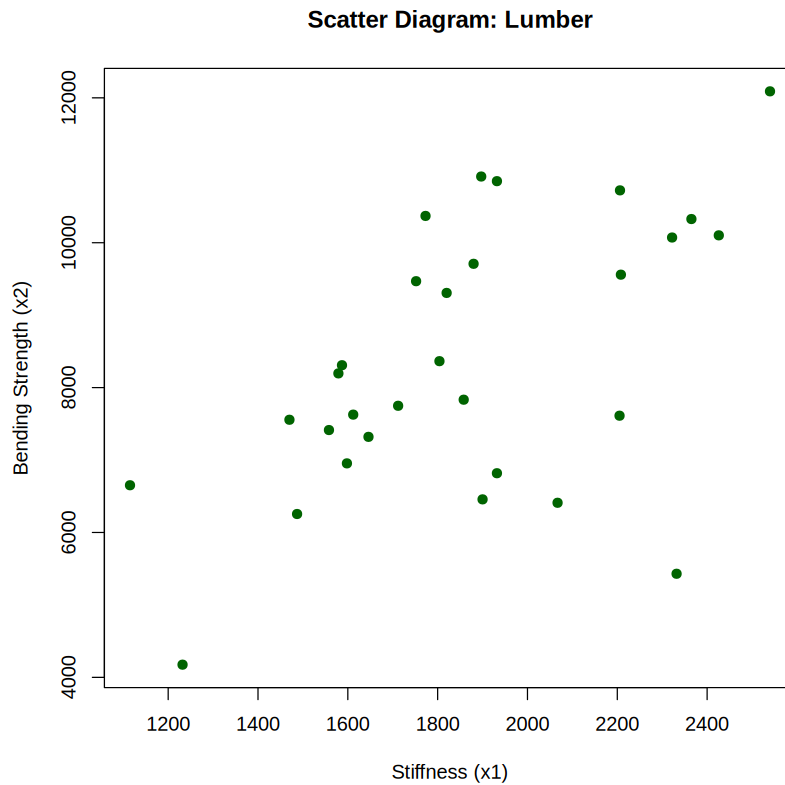
```
In [11... d2_lumber <- mahalnobis(lumber_data, xbar, S)
```

```
qqplot(qchisq(ppoints(n), df = p), d2_lumber,
       main = "Chi-Square Q-Q Plot (Lumber)", xlab="Chi-Square Quantile",
       abline(a = 0, b = 1, col="red", lwd=2))

plot(lumber_data$stiffness, lumber_data$bending_strength,
     main="Scatter Diagram: Lumber",
     xlab="Stiffness (x1)", ylab="Bending Strength (x2)", pch=19, col="da
```

Chi-Square Q-Q Plot (Lumber)





Q-Q Plot: The plotted points align very closely to the red reference line. This shows that the quantiles of the sample distances follow the theoretical quantiles of the Chi-Square distribution, meaning that there is strong evidence that the underlying population is multivariate normal.

Scatter Diagram: The scatter plot of stiffness versus bending strength shows a scattered, upward-sloping elliptical cloud of points. Unlike the previous dataset, there is no extreme outlier that skews the data.

Thus it can be said that, the bivariate normal distribution is a viable population model for the lumber dataset.